

Wavelet Practical

STA5090Z: Topics in Advanced Regression

April 2025

Question 1: Daubechies wavelets

Ingrid Daubechies developed a family of compactly supported wavelets. The Daubechies family wavelets are named according to their filter numbers. Filter number 1 is the Haar wavelet.

For this question, use the `wavethresh` package (Nason, 2016).

Plot the Daubechies mother wavelets for filter numbers 1 to 10.

```
draw.default(filter.number = ..., family = "DaubExPhase")
```

There is no closed-form expression for most wavelets. In practice they are numerically approximated. These approximations can be derived because wavelets are constructed so that they follow certain mathematical conditions, e.g. integrate to 0 and have a certain number of vanishing moments. Software for wavelets often stores these approximations of filter coefficients, and these can then be used to plot the wavelet.

For example, the filter coefficients for the Daubechies Extremal Phase wavelet with 4 vanishing moments are:

```
library(wavethresh)
(fc <- filter.select(filter.number = 4, family = "DaubExPhase"))
```

```
$H
[1] 0.23037781 0.71484657 0.63088077 -0.02798377 -0.18703481 0.03084138
[7] 0.03288301 -0.01059740
```

```
$G
NULL
```

```
$name
```

```
[1] "Daub cmpct on ext. phase N=4"
```

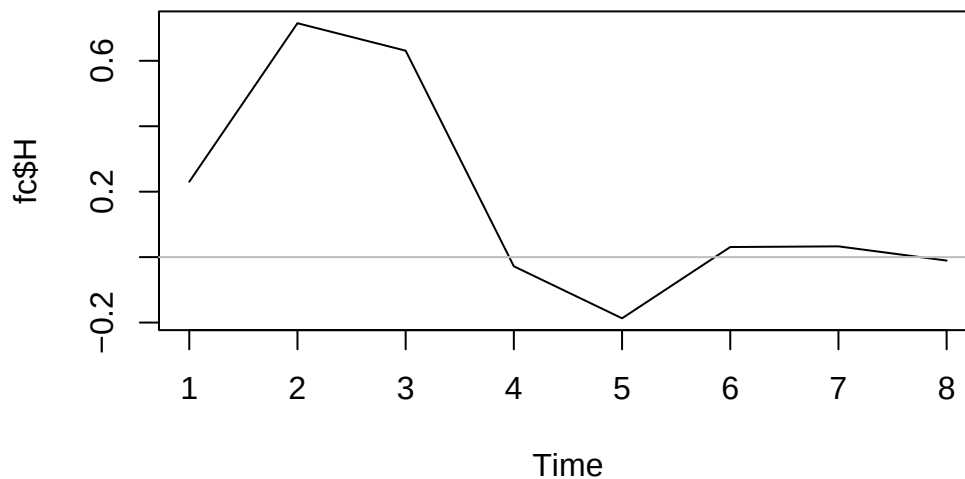
```
$family
```

```
[1] "DaubExPhase"
```

```
$filter.number
```

```
[1] 4
```

```
plot.ts(fc$H)
abline(h = 0, col = "grey")
```



If this wavelet is stretched over a wider x-range the coefficients for each value of x are found through a recursive algorithm (Rowe and Abbott, 1995).

Question 2: Mexican hat wavelet

The Mexican Hat mother wavelet function is:

$$\psi(t) = (1 - t^2) \exp(-t^2/2)$$

How would you translate and dilate this to obtain daughter wavelets?

Question 3: Pyramid algorithm

Here is the simple vector of values used in the slides for the pyramidal algorithm:

```
y <- c(1, 1, 7, 9, 2, 8, 8, 6)
```

Use function `wd()` to perform a *discrete wavelet transform*, using Haar wavelets. Again you need to specify filter number and family. A plot of this object returns the detail coefficients. To better see the coefficients at finer scales you can add the option `scaling = "by.level"`. Usually the coefficients at finer scales are smaller.

The detail coefficients are saved in `D`, but in a long vector. To extract them use function `accessD()`. How do they relate to the values in the slides? When you plot the coefficients from `C` and `D` you obtain the smoothed function (approximation) and the detail for that level (resolution).

You can see the basis matrix W using function `GenW()`. Use this to obtain the wavelet coefficients θ . And again, having obtained θ , get back to the original vector y (wavelet transform and inverse wavelet transform).

The signal can also be reconstructed using function `wr()`, either given the wavelet coefficients (or a discrete wavelet transform object) or some shrunken version of this.

Question 4: Nuclear magnetic resonance

The NMR (nuclear magnetic resonance) signal data (ESL pg. 177) can be found here: <http://www.isnld.com/IIICH05/RaphNMR.asc>. This is a signal for which wavelets should work very well, some sections are smooth (apart from noise), but there are occasional sudden spikes in the data.

Do a discrete wavelet transform to obtain the wavelet coefficients. The symmlet-8 wavelet (used in ESL) is a LEAST ASYMMETRIC wavelet, family = "DaubLeAsymm", filter number 8.

Smooth the signal using different shrinkage methods (universal, SURE), plot on top of original signal, see function `threshold()`.

As a comparison, fit a spline to these data and add to the plot. How well does the spline work here?

To do something similar to ESL, you can use package `wmtsa` (William Constantine and Donald Percival (2017). `wmtsa`: Wavelet Methods for Time Series Analysis. R package version 2.0-3. <https://CRAN.R-project.org/package=wmtsa>)

- discrete wavelet transform: `wavDWT`
- shrinking and reconstruction: `wavShrink`
- continuous wavelet transform: `wavCWT`

Maximal Overlap DWT

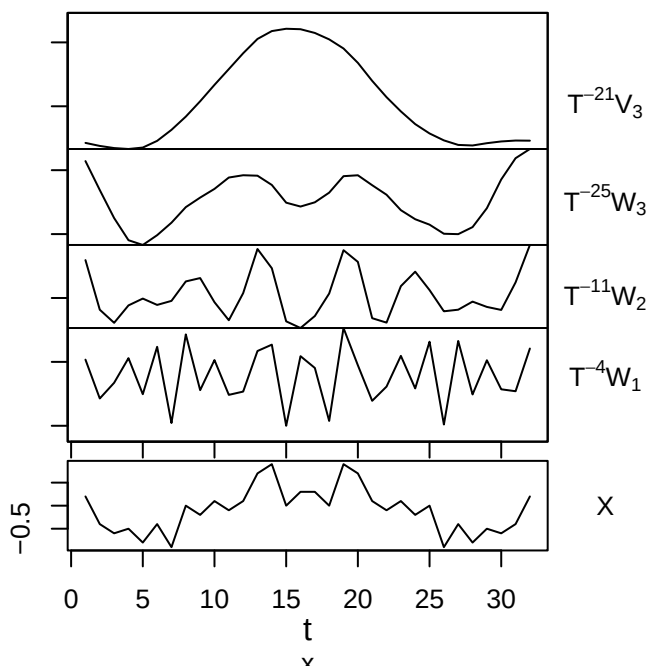
```
library(wavelets)

# obtain the two series listed in Percival and Walden (2000), page 42
X1 <- c(.2,-.4,-.6,-.5,-.8,-.4,-.9,0,-.2,.1,-.1,.1,.7,.9,0,.3)
X2 <- c(.2,-.4,-.6,-.5,-.8,-.4,-.9,0,-.2,.1,-.1,.1,-.7,.9,0,.3)

# combine them and compute modDWT
newX <- cbind(X1,X2)
length(newX)
```

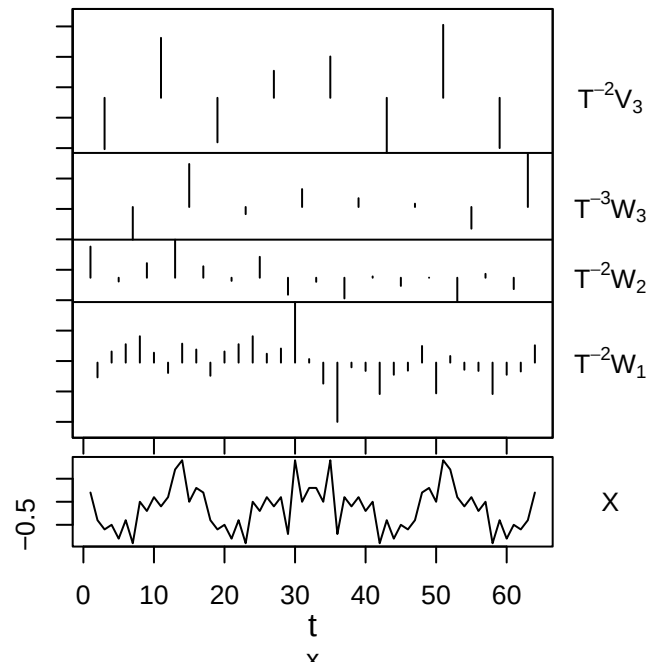
[1] 32

```
wtmo <- modwt(newX, n.level=3, boundary="reflection", fast=FALSE)
#str(wtmo)
plot(wtmo)
```



```
# wtmo@W
# DWT
```

```
newX <- cbind(X1,X2)
wt <- dwt(newX, n.levels=3, boundary="reflection", fast=FALSE)
plot(wt)
```



References

- Nason G (2022). *wavethresh: Wavelets Statistics and Transforms*. R package version 4.7.2, <https://CRAN.R-project.org/package=wavethresh>.
- Rowe ACH, Abbott PC. 1995. Daubechies wavelets and Mathematica. *Comput Phys.* 9(6):635.